Experiment 8: Conduct Z test for the Data Given using Python

Objective: To test whether the average weight of a species of birds differs from 150 grams.

Procedure: 1. Null Hypothesis (H_0): The average weight of the birds is 150 grams.

2. Alternative Hypothesis (H_1): The average weight of the birds is not 150 grams.

3. Sample: Measure the weights of 30 birds randomly selected from the population.

4. Z-Test: Conduct a Z-test to compare the sample mean to 150 grams.

5. Decision Rule: Use a significance level of α = 0.05.

PROGRAM:

```
import numpy as np
from scipy.stats import norm
# Sample data
data = np.array([64, 68, 61, 65, 63, 67, 62, 66, 60, 64])
# Hypothesized population mean
population_mean = 64
# Known population standard deviation
population_std = 5
# Sample size
n = len(data)
# Calculate sample mean
sample_mean = np.mean(data)
# Calculate Z-score
z_score = (sample_mean - population_mean) / (population_std / np.sqrt(n))
# Calculate p-value
p_value = 2 * (1 - norm.cdf(abs(z_score)))
# Significance level
alpha = 0.05
# Display results
print("Sample Mean:", sample_mean)
print("Z-Score:", z_score)
print("P-Value:", p_value)
# Decision
if p_value < alpha:
```

```
    print("Reject the null hypothesis.")
else:
    print("Fail to reject the null hypothesis.")
```

OUTPUT:

Sample Mean: 64.0

Z-Score: 0.0

P-Value: 1.0

Fail to reject the null hypothesis.